

SHILPA KALA – DIGITAL PORTFOLIO ASSISTANT

Detailed Project Report

Developed Using: Android Studio & Kotlin

Project Domain: Self Employment / Digital Portfolio

Platform: Android Mobile Application

GitHub Repository:

<https://github.com/Bhargavigo/ShilpaKala.android.app>

Abstract

Shilpa Kala is an Android-based mobile application developed to support artisans and self-employed craftsmen in creating professional digital product portfolios. The application enables users to capture product images, add branding and product details, save processed images, and share them through social media platforms. The project improves digital visibility and product presentation for handmade products.

1. Introduction

In today's digital world, product presentation plays an important role in attracting customers. Many artisans create high-quality handmade products but struggle to present them professionally online due to lack of photography and editing tools. Shilpa Kala was developed to solve this issue by providing a simple mobile platform for capturing, branding, and sharing artisan product images.

2. Problem Statement

Traditional artisans face several problems while marketing products online. Product images often appear simple and unprofessional, reducing their value in digital platforms. Most artisans do not have access to professional editing software or photography tools. The application solves this problem through guided image capture, branding overlays, product detail integration, and social media sharing.

3. Objectives

- Develop an Android application for artisan portfolio creation
- Capture product images using CameraX API
- Add branding and product information to images
- Save processed images in gallery
- Share images through social media applications
- Improve digital product presentation and self-employment opportunities

4. Scope of the Project

The project focuses on providing a digital portfolio solution for artisans. The application supports image capture, branding, gallery management, image storage, and sharing. The system is designed for Android smartphones and can be expanded in future with cloud storage and AI image enhancement features.

5. Technologies Used

- Android Studio – Application development environment
- Kotlin – Programming language
- XML – User interface design
- CameraX API – Camera integration
- Bitmap & Canvas – Image processing
- File Storage API – Saving images to gallery
- GitHub – Version control and project hosting

6. System Architecture

The application follows a modular multi-screen architecture: Home Screen → Camera Screen → Details Screen → Image Processing → Save Image → Gallery Screen → Share Image. Each module performs a specific task and works together to provide a smooth workflow.

7. Methodology

1. User opens the application.
2. Product image is captured using CameraX.
3. User enters product details such as product name, wood type, and price.
4. Bitmap image processing is applied.
5. Branding and overlay text are added using Canvas.
6. Final image is saved to gallery.
7. User can view and share the image through social media.

8. Modules Description

Home Module: Provides navigation to camera and gallery.

Camera Module: Captures product images using CameraX API.

Details Module: Collects product details from the user.

Image Processing Module: Adds branding, overlays, and price tags.

Gallery Module: Displays saved images in grid format.

Sharing Module: Shares images through WhatsApp and social media apps.

9. Algorithms Used

Bitmap Processing Algorithm: Loads and edits captured images.

Canvas Drawing Algorithm: Draws product details and branding overlays on images.

File Handling Algorithm: Creates folders, saves images, and refreshes gallery storage.

10. Testing and Optimization

The application was tested for camera functionality, gallery loading, image saving, social sharing, and UI responsiveness. Optimization was performed to improve image loading performance, navigation flow, and text readability on images.

11. Output and Results

The application successfully generated professional-looking portfolio images with branding, stylish overlays, and price tags. Users were able to save images locally, view them in gallery format, and share them through social

media applications.

12. Advantages

- Easy to use mobile application
- Professional product presentation
- Supports self-employment and artisans
- Fast image branding process
- Social media sharing support
- Attractive user interface

13. Limitations

- Works only on Android devices
- No cloud storage integration currently
- Depends on mobile camera quality
- Limited advanced image editing features

14. Future Scope

Future enhancements may include cloud storage integration, AI-based image enhancement, online artisan portfolio management, e-commerce integration, multi-language support, and advanced product catalog systems.

15. Conclusion

The Shilpa Kala project successfully demonstrates how Android technology can support artisans and self-employed craftsmen through digital portfolio creation. The application improves product presentation, branding, and visibility while providing a user-friendly and professional solution for image management and sharing.

16. Technology Summary Table

Technology	Purpose
Android Studio	Application Development
Kotlin	Programming Language
XML	User Interface Design
CameraX API	Camera Integration
Bitmap & Canvas	Image Processing
GitHub	Version Control